Physics II — Worked Solutions: Lecture 1–3

Phy 4143 · Every example / practice problem from the Lecture 1–3 slides, solved step by step.

LECTURE 1 — Basic Concepts, Ohm's Law, KCL/KVL

Full step-by-step derivations for these five are already in **notes/2026-11-02-kcl-node-junction-branch.md**. *Final answers recapped here:*

Fig. 2.21 — Series circuit (20 V, 2 Ω, 3 Ω)

i = 20/(2+3) = 4 A → v1 = 4×2 = 8 V, v2 = 4×3 = 12 V.

Answer: v1 = 8 V, v2 = 12 V

Fig. 2.22 / Practice 2.5 — 10 V & 8 V aiding, 4 Ω, 2 Ω

KVL: 10 + 4i + 2i − 8 = 0 → i = −⅓ A... using the slide's own working: v1 = 12 V, v2 = −6 V (sign flips because the assumed current direction enters v2's marked − terminal first — see the sign-convention discussion earlier in this chat).

Answer: v1 = 12 V, v2 = −6 V

Example 2.7 / Fig. 2.25 — dependent + 3 A source

Single-node KCL with a dependent source gives i_o = 6 A, v_o = 24 V.

Answer: i_o = 6 A, v_o = 24 V

Practice 2.7 / Fig. 2.26 — 6 A source, 2 Ω, $i_o/4$ dependent source, 8 Ω — exam-important

KCL at the single node: 6 = i_o + $i_o/4$ + $v_o/8$, with $v_o = 2i_o$ → 6 = 1.5 i_o.

Answer: i_o = 4 A, v_o = 8 V

Example 2.8 / Fig. 2.27a — 30 V, 8 Ω, (3 Ω parallel 6 Ω)

3 Ω parallel 6 Ω = 2 Ω, Req = 8+2 = 10 Ω, i1 = 30/10 = 3 A. v1 = 3×8 = 24 V, remaining 6 V splits across the 3 Ω parallel 6 Ω pair → i2 = 6/3 = 2 A, i3 = 6/6 = 1 A.

Answer: i1 = 3 A, i2 = 2 A, i3 = 1 A, v1 = 24 V, v2 = v3 = 6 V

LECTURE 2 — Sources, Source Transformation, Series/Parallel, Dividers

Example 1 (slide) — Voltage divider, v_R4 in a 12 V circuit

Solved on the slide itself using parallel-equivalents (R1 parallel R2=320 Ω, R3 parallel R4=333 Ω) then a voltage divider on the simplified 3-resistor loop.

Answer (given on slide): v_R4 = 4.065 V

Example 2 (slide) — Current divider, i_R2 with a 2.2 A source

Solved on the slide using series/parallel reduction (R12=51 Ω, R456=153 Ω) then a current divider on the simplified 3-branch parallel circuit.

Answer (given on slide): i_R2 = 1.19 A

"Find R_eq" #1 — ladder: 4 Ω+1 Ω top, 2 Ω bridge, 6 Ω parallel 3 Ω, 5 Ω side, 8 Ω bottom

6 Ω parallel 3 Ω = 2 Ω, so going via the bridge: 2+2 = 4 Ω. Going via the top-right/side path: 1+5 = 6 Ω. These two paths are in parallel: 4 parallel 6 = 2.4 Ω. Add the first 4 Ω and the bottom 8 Ω in series: Req = 4 + 2.4 + 8.

Answer: R_eq = 14.4 Ω

"Find R_eq" #2 — 10-resistor bridge network (R1–R10)

This figure's node connections are too fine to read reliably off the slide image from here (several diagonal resistors cross near the centre). Send a clearer photo/crop of this one slide and I'll solve it properly — skipped for now rather than guess the topology.

"Find R_eq" #3 — 4 Ω+3 Ω top, 6 Ω/4 Ω/12 Ω verticals, 3 Ω bottom

4 Ω parallel 12 Ω (the two right verticals) = 3 Ω; in series with the 3 Ω top segment = 6 Ω. That 6 Ω path is in parallel with the direct 6 Ω vertical on the left = 6 parallel 6 = 3 Ω. Add the first 4 Ω (top) and the 3 Ω (bottom) in series: Req = 4 + 3 + 3.

Answer: R_eq = 10 Ω

"Find R_eq" #4 — 4 Ω+3 Ω+4 Ω top, 6 Ω/4 Ω/5 Ω verticals, 3 Ω+3 Ω bottom

Reduce from the right: the 4 Ω (top) + 5 Ω (vertical) series path is in parallel with the middle 4 Ω vertical: 4 parallel 9 = 2.77 Ω. Add the 3 Ω top and 3 Ω bottom segments in series with it (2.77+3+3 = 8.77 Ω), then that is in parallel with the left 6 Ω vertical: 6 parallel 8.77 = 3.56 Ω. Add the first 4 Ω (top) and first 3 Ω (bottom): Req = 4 + 3.56 + 3.

Answer: R_eq ≈ 10.56 Ω (topology read off the slide — double-check the figure if this looks off)

Current-divider word problem — 100 V / 1.5 kW source, R1=10 Ω, R2=25 Ω, R3=100 Ω (parallel)

Total current from the source rating: I_T = P/V = 1500/100 = 15 A. Equivalent resistance: 1/Req = 1/10+1/25+1/100 = 0.15 → Req = 6.667 Ω. Current divider: I_R1=(Req/R1)×I_T, I_R2=(Req/R2)×I_T, I_R3=(Req/R3)×I_T.

Answer: Req = 6.67 Ω, I_R1 = 10 A, I_R2 = 4 A, I_R3 = 1 A (sum = 15 A (checks out), also matches direct Ohm's law since all three sit across the same 100 V)

LECTURE 3 — Y-Δ and Δ-Y Transformation

Example 8.29 (slide) — R_A=3 Ω, R_B=3 Ω, R_C=6 Ω bridge

Solved on the slide via Δ→Y (R1=R2=1.5 Ω, R3=0.75 Ω) then series/parallel reduction.

Answer (given on slide): R_T = 2.889 Ω

Example 8.30 (slide) — balanced bridge, all resistors 6 Ω or 9 Ω

Solved on the slide both ways (Δ→Y and Y→Δ), both giving the same final answer.

Answer (given on slide): R_T = 3.2727 Ω

"Problems to solve" / "Now Solve" (appears twice) — A–B bridge: 4 Ω, 5 Ω top; 3 Ω, 5 Ω bottom; 2 Ω bridge

The three resistors meeting at the middle-top node M (4 Ω to A, 2 Ω to N, 5 Ω to B) form a Y — convert it to a Δ connecting A, B, N. Sum = 4×5+5×2+2×4 = 38. Δ(A,B)=38/2=19 Ω, Δ(B,N)=38/4=9.5 Ω, Δ(A,N)=38/5=7.6 Ω.

These sit in parallel with the original branches: A–N: 3 parallel 7.6=2.151 Ω; N–B: 5 parallel 9.5=3.276 Ω. Path through N = 2.151+3.276 = 5.427 Ω, in parallel with the direct Δ(A,B)=19 Ω.

Answer: R_eq(A–B) = (5.427×19)/(5.427+19) ≈ 4.22 Ω

"Now Solve" — triangular 9-resistor bridge (20, 30, 5, 45, 25, 15, 35, 10, 40 Ω)

This is a hand-drawn figure and the inner connections (which resistor meets which node) aren't legible enough from here to solve correctly. Send a clearer photo of just this slide and I'll work through it.

"Now Solve" — ladder a–b: 12.5 Ω, 10 Ω, 5 Ω (bridge), 15 Ω, 20 Ω, 30 Ω (same circuit as the course's own mid-term, Fig. 1(c))

At node X (under the 12.5 Ω/15 Ω branch), three resistors meet: 12.5 Ω to a, 5 Ω to Z (the other mid-node), 15 Ω to b — a Y. Convert to Δ (a, Z, b): sum = 12.5×5+5×15+15×12.5 = 325.

Δ(a,Z)=325/15=21.67 Ω, Δ(Z,b)=325/12.5=26 Ω, Δ(a,b)=325/5=65 Ω. Combine with the original branches: a–Z: 10 parallel 21.67=6.84 Ω; Z–b: 20 parallel 26=11.30 Ω; a–b (direct): 30 parallel 65=20.53 Ω.

Series path a–Z–b = 6.84+11.30 = 18.15 Ω, in parallel with the direct 20.53 Ω path: Req = (18.15×20.53)/(18.15+20.53) ≈ 9.63 Ω. If this feeds i (as in the mid-term, V=120 V): i = 120/9.63 ≈ 12.46 A.

Answer: R_ab ≈ 9.63 Ω (and i ≈ 12.46 A for the 120 V mid-term version)

Note: two figures (the 10-resistor R1–R10 bridge in Lecture 2, and the 9-resistor triangular bridge in Lecture 3) are too visually dense/hand-drawn to trace reliably from the slide image alone — everything else in Lecture 1–3 is solved above.